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GRADES 7–10

EMERGENCY SUB PLANS PACK

for DIGITAL TECHNOLOGIES

★ *Keep learning on track—even when plans change.* ★

- NO-PREP ACTIVITIES
- READY-TO-USE RESOURCES
- PERFECT FOR EMERGENCY LESSONS
- GREAT FOR SUBSTITUTE TEACHERS
- ENGAGING DIGITAL TECH TASKS
- PRINT OR DIGITAL USE

YOUR **LIFESAVER** IN THE CLASSROOM

SAVE TIME | REDUCE STRESS | KEEP STUDENTS ENGAGED

MADE BY TEACHERS FOR TEACHERS

BEFORE YOU NEED IT

Set this up while you are well

FOR THE REGULAR TEACHER

15 MINUTES

You are reading this because you are **well**. Spend fifteen minutes now and this pack will run itself on a day when you are not.

The one job you have today

Print pages 3–6 of this pack and leave them in a clearly labeled folder where your substitute teacher will find them — top drawer, on the desk, or with the front office. Everything a substitute needs is in those four pages.

You do not need to print the lessons in advance. The substitute teacher chooses one on the day and photocopies it.

SETTING IT UP · FIFTEEN MINUTES, ONCE

1. **Print the whole pack once.** Put it in a display folder or bind it. Write your name and room number on the cover.
2. **Print pages 3–6 a second time** and clip them to the front. These are the pages a substitute teacher reads first.
3. **Tell your department head where it lives.** On a genuine emergency day you will not be answering your phone, so somebody else needs to know.
4. **Cross off the lessons you have already used.** There is a tracker on page 4. Two years of substitute days will not repeat if you keep it current.
5. **Optional: leave a class list** and a note about any students who need specific support, seating or a reader.

Why these lessons are unplugged

On a substitute day the laptop cart is often locked, the login server is down, or nobody knows the room password. Every lesson here runs on paper.

Each lesson also has an **If the computers are working** note, so the same sheet still works when technology cooperates.

Why they are not busy work

Every lesson targets a genuine Digital Technologies concept and produces marked evidence you can put in a folder.

If you are away for a week, running four of these in sequence is a defensible unit of work, not a holding pattern.

A word about the answer keys

Every answer key sits on the **last page of its lesson**. If you hand a lesson to a substitute teacher, hand them the answer key too — it is the single thing that lets a non-specialist mark with confidence and answer a student who says *“but why?”*

This is Volume One

Volume 1 is the foundation pack: algorithms, binary, tracing, flowcharts, security, networks, efficiency, debugging, AI and privacy. Volumes 2, 3 and 4 add thirty further lessons and none of them repeat anything here.

SUBSTITUTE TEACHER · START HERE

You do not need to know this subject

READ THIS FIRST

2 MINUTES

You have walked into a Digital Technologies class. You may have never taught the subject. That is completely fine — this pack was written for you.

Read this and you are ready

You do not need to know how to code. Every lesson is a paper worksheet with a full answer key. Your job is to hand it out, keep the room settled, and check work as students finish. You are not expected to teach the content.

DO THESE FOUR THINGS

1. **Pick one lesson** from the table on the next page. Any of them works for any class from Grade 7 to Grade 10. If you cannot decide, use Lesson 05.
2. **Photocopy the student pages only.** Each lesson tells you exactly which page numbers to copy. One copy per student.
3. **Keep the teacher page and the answer key.** Do not photocopy those.
4. **Read the teacher page once** before the bell. It takes about two minutes and includes the exact words you can say to start the lesson.

If a student says “we have done this”

Ask them to prove it by completing the extension section at the end. If they genuinely have, switch them to a different lesson from the table — the pack has ten, and they will not have done them all.

If you finish early

Every lesson has an extension task on its final student page. Point students there. There is also a discussion prompt on the teacher page you can run as a whole class for the last five minutes.

What to leave for the regular teacher

Collect the completed sheets. Fill in the handover note on the last page of this pack — it takes ninety seconds and tells the regular teacher which lesson you used, how far the class got, and anything that happened.

Leave both on the desk.

SUBSTITUTE TEACHER · STEP ONE

Choose your lesson

PICK ANY ONE

10 OPTIONS

Every lesson is standalone. There is no order and no prerequisite — pick whichever suits the class in front of you.

#	LESSON	WHAT IT COVERS	DIFFICULTY	MINS	STUDENT PAGES
01	Follow My Instructions Exactly	Algorithms and precision	★	60	7–9
02	Counting Like a Computer	Binary and data representation	★★	60	12–14
03	Trace the Code	Reading programs by hand	★★	55	17–19
04	Flowchart Detective	Logic and decisions	★	55	22–24
05	How Attacks Actually Work	Cyber security	★	60	27–29
06	Sending a Message Across the World	Networks and the internet	★★	60	32–34
07	Sorting and Searching by Hand	Algorithm efficiency	★★★	60	37–39
08	Bug Hunt	Debugging and error types	★★★	55	42–44
09	What AI Can and Cannot Do	Artificial intelligence and ethics	★	60	47–49
10	Your Digital Footprint	Privacy and digital citizenship	★	55	52–54

Easiest to supervise

05, 09 and 10. Scenario and discussion based. Students have opinions, so the room stays busy and nobody gets stuck.

Best for a quiet, working class

02, 03 and 07. Puzzle-like with definite answers. Strong students will race; that is what the extensions are for.

Best for a restless class

01 and 04. Partner activities with movement and a bit of humor built in. Noisier, but productively so.

Used-lesson tracker · for the regular teacher

LESSON	DATE USED	CLASS	NOTES
01 — Instructions			
02 — Binary			
03 — Trace the code			
04 — Flowcharts			
05 — Cyber security			
06 — Networks			
07 — Sorting			
08 — Bug Hunt			
09 — AI			
10 — Digital Footprint			

SUBSTITUTE TEACHER · STEP TWO

The first ten minutes

WHAT TO SAY

SCRIPT

The first four minutes decide the next fifty. Here is a script that works even if you have never met this class.

SAY THIS AT THE DOOR

> *"Come in, sit down, and leave your bags off the desks. You will not need a computer today. Everything is on paper."*

THEN, ONCE THEY ARE SEATED

> *"Your teacher is away. My name is _____. We are doing a proper Digital Technologies lesson today, not a free period. There is a worksheet, it gets collected at the end, and your teacher will see it."*

Why that last sentence matters

The single most useful thing you can say is that the work will be seen by their regular teacher. It is also true — there is a handover note at the back of this pack.

THE SHAPE OF EVERY LESSON IN THIS PACK

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Settle, hand out sheets	Use the script above
5–10	Read the opening page together	Ask one student to read it aloud
10–45	Students work through the tasks	Circulate. Check answers against the key
45–55	Extension task for those who finish	Point them to the last student page
55–60	Collect sheets, pack up	Fill in the handover note

If a student refuses to work

Offer the smaller ask: *"Just do question one and show me."* Most refusals are about not knowing where to start, not defiance.

Note it on the handover sheet. Do not escalate a single lesson into a battle you will not be there to finish.

If the whole class claims they cannot do it

Do the first question together on the board. Every lesson has a worked example on the first student page.

If it is genuinely too hard for the grade level, switch to Lesson 05 or Lesson 10 — both need no prior knowledge at all.

Photocopying reminder

Student pages only. Each lesson's teacher page states the exact page range. The **answer key is always the last page** of the lesson — keep it.

LESSON 01 · TEACHER BRIEFING

Follow My Instructions Exactly

READ BEFORE THE BELL

2 MINUTES

Students discover that a computer does precisely what it is told — which is rarely what you meant. It is the foundation of every algorithm they will ever write, and it is genuinely funny when it goes wrong.

You do not need to know this subject

This lesson is about writing clear instructions in plain English. There is no code and no technical vocabulary you need to hold. If you can tell whether a sentence is vague, you can run this lesson.

Photocopy these pages only

Pages 7–9

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Nothing else.

WHAT TO SAY TO START

> *"Today you are going to find out how annoying a computer actually is. It will do exactly what you say. Not what you meant — what you said. By the end you will understand why programmers spend so much time being precise."*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the opening panel together	Have a student read 'the obedient idiot' aloud
12–22	Tasks 1.1 and 1.2 — spotting vagueness	Circulate. Task 1.1 has no single right drawing
22–38	Task 1.3 — write and swap the sandwich algorithm	This is the noisy part. Let it be noisy
38–50	Tasks 1.4 and 1.5 — loops and decisions	Answer key has model answers
50–60	Collect sheets, run the discussion prompt	Prompt is at the foot of this page

Questions students will ask

"Is my drawing wrong?"
No. That is the point of Task 1.1. The instructions do not say how big or which direction, so several different drawings are all correct.

"How many steps should the sandwich have?"
At least eight. If a student writes four, ask them what a robot would do with step one.

"Can I write 'spread the peanut butter'?"
Ask: with what? Which side? How much? That question is the whole lesson.

If they finish early

Task 1.5 on the third student page is the extension — writing instructions for a robot that has to make a decision and repeat itself. Strong students can then write instructions for tying a shoelace, which is far harder than it sounds.

If the computers are working

Students can type their sandwich algorithm and have a classmate follow it literally on screen, or try it in any block-based coding tool where the computer really will refuse a vague instruction.

Curriculum focus · for the regular teacher's records

Designing algorithms; representing a solution as a sequence of steps; using branching and iteration in a plain-language algorithm. Discussion prompt for the last five minutes: *"A human can follow 'add a little salt'. Why can a computer never follow it — and is that a weakness or a strength?"*

LESSON 01 · FOLLOW MY INSTRUCTIONS EXACTLY

The obedient idiot

UNDERSTAND

12 MIN

NAME	CLASS	DATE

Computers are not clever. They are fast and obedient.

A computer will carry out your instructions perfectly, in order, without ever stopping to ask what you meant. That sounds helpful. In practice it means a single vague word can ruin an entire program.

Humans fill in gaps automatically. If someone says “grab me a drink” you work out the rest. A computer has no idea what “a drink” is, where it is, or what “grab” means.

Task 1.1 · Draw exactly what it says

DO

Follow these five instructions in the box. Do not use common sense. Do only what the words actually say.

instructions — FIVE STEPS

1. Draw a dot near the middle of the box.
2. Draw a straight line starting at the dot.
3. From the end of that line, draw another line.
4. Join the end of the second line back to the dot.
5. Shade in the shape you have made.

Now compare with the person next to you. Your drawings are almost certainly different. List **four** things the instructions never told you:

Vague words and where they hide

KEY SKILL

20 MIN

Task 1.2 · Find the vague word

PROVE

Each instruction below would confuse a computer. Circle the word or phrase that is vague, then rewrite the instruction so it could not be misunderstood.

#	INSTRUCTION	REWRITE IT PRECISELY
a	Add a little salt.	
b	Move the mouse to the button and click it.	
c	Sort the list.	
d	Wait for a while, then try again.	
e	Keep going until it looks right.	

Task 1.3 · The sandwich algorithm

WRITE + TEST

Write instructions for making a peanut butter and jelly sandwich. Assume the reader is a robot that has never seen food. Use at least **eight numbered steps**.

Swap with a partner. Read their instructions out loud and act them out *literally*. Then answer:

The first step that failed was step _____ because _____

Repeating and deciding

STRETCH · 15 MIN

Two words that make instructions powerful

REPEAT saves you writing the same line over and over.

repeat — PLAIN ENGLISH

```
REPEAT 4 TIMES:
    put one slice in the toaster
    press the lever down
```

IF lets the instructions handle more than one situation.

if — PLAIN ENGLISH

```
IF the toast is burnt:
    throw it away
OTHERWISE:
    butter it
```

Task 1.4 · Rewrite using REPEAT

DO

six lines — BEFORE

```
water plant 1
water plant 2
water plant 3
water plant 4
water plant 5
water plant 6
```

A robot must water six plants. Rewrite these six identical lines using REPEAT.

Task 1.5 · The plant-watering robot

EXTENSION

Write instructions for a robot that looks after one plant for seven days. It must check the soil each day and water it **only if** the soil is dry. Use both REPEAT and IF.

Task 1.6 · The last word

THINK

A human can follow “add a little salt”. A computer never can. Why is that actually useful, not just annoying?

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Follow My Instructions Exactly

NOT FOR STUDENTS

ANSWERS

How to grade this lesson

Most answers here are **judgment calls, not exact matches**. Give the credit if the student has removed the ambiguity, even if their wording differs from the model answer below.

Task 1.1 · what the instructions never said

Any four of: how big the dot is · how long each line is · in which direction each line goes · whether the lines are straight · what "near the middle" means · what color to shade · whether the shape is closed at all.

All differing drawings are correct — that is the point.

Task 1.2 · vague words

#	VAGUE PART	MODEL REWRITE
a	a little	Add 1/2 teaspoon of salt.
b	it	Move the pointer over the OK button, then press the left mouse button once.
c	sort	Arrange the list from smallest to largest.
d	a while	Wait 30 seconds, then try again.
e	looks right	Repeat until the number reaches 100.

Task 1.3 · sandwich algorithm

Full credit for eight or more steps that each contain a single action. Look for: which item is picked up, which surface, which side, how much, and what to do with the knife. Almost every class forgets to **open the jar** and to **take the bread out of the bag**. Those are the best teaching moments — point them out rather than grading them wrong.

Task 1.4 · model answer

after — PLAIN ENGLISH

```
REPEAT 6 TIMES:
    water the next plant
```

Accept any version that repeats once and moves to the next plant each time.

Task 1.5 · model answer

seven days — PLAIN ENGLISH

```
REPEAT 7 TIMES:
    check the soil
    IF the soil is dry:
        pour 1 cup of water on it
    OTHERWISE:
        do nothing
    wait 24 hours
```

The point of the last question

A computer's refusal to guess is what makes it **reliable and repeatable**. A human cook adds a different amount of salt every time; a machine adds exactly half a teaspoon, a million times, identically. Accept any answer that reaches consistency, reliability or repeatability.

LESSON 02 · TEACHER BRIEFING

Counting Like a Computer

READ BEFORE THE BELL

2 MINUTES

Students learn to read and write binary numbers, decode a secret message in ASCII, and work out real file sizes. It is arithmetic with a puzzle feel, and it produces very gradeable work.

You do not need to know this subject

Every answer in this lesson is a definite number and every one of them is in the answer key. You do not need to be able to do binary yourself — you need the key, which is the last page of this lesson.

Photocopy these pages only

Pages 12–14

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. A calculator is useful for the last section but not essential.

WHAT TO SAY TO START

> *"Computers cannot count to ten. They only have two digits — zero and one — because a wire is either on or off. Today you will learn to count the way a machine does, and then decode a message written in it."*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the panel and worked example together	Do the number 13 on the board with them
12–26	Tasks 2.1 and 2.2 — converting both ways	Every answer is in the key
26–38	Tasks 2.3 and 2.4 — the secret message	The message decodes to a real word
38–50	Tasks 2.5 and 2.6 — file sizes	Calculator allowed if they have one
50–60	Extension, collect sheets	Discussion prompt below

Questions students will ask

"Why do we even need binary?"
A wire can be on or off, and nothing in between. Two states means two digits. Everything a computer stores has to be built from that.

"Is the answer 8 or 1000?"
Both — they are the same number written two ways, the same way 'twelve' and '12' are. That confusion is normal and worth saying out loud.

"Can I use a calculator?"
For the file-size questions, yes. For the binary conversions, no — the method is the point.

If they finish early

Task 2.4 asks students to write their own initials in binary. Beyond that, ask them to work out the largest number that can be stored in 16 bits, and why a very old game console could only show 256 colors.

If the computers are working

Students can check their conversions with the programmer mode of any calculator app, or search for a binary converter. Ask them to convert first, then check — not the other way round.

Curriculum focus · for the regular teacher's records

Representing data in binary; how digital systems represent text, numbers and images; units of data storage. Discussion prompt for the last five minutes: *"A photo on your phone is about 4 MB. That is over thirty million ones and zeros. Where are they all kept?"*

LESSON 02 · COUNTING LIKE A COMPUTER

Counting with only two digits

UNDERSTAND

14 MIN

NAME _____ CLASS _____ DATE _____

A wire is either on or off. That is all a computer has.

We count in **tens** because we have ten fingers. Each column is worth ten times the one to its right: 1, 10, 100, 1000.

A computer counts in **twos**. Each column is worth double the one to its right: 1, 2, 4, 8, 16, 32, 64, 128. A column is either used (1) or not used (0).

Worked example · write 13 in binary

Work from the left. Ask each column: does it fit into what is left over?

128	64	32	16	8	4	2	1
0	0	0	0	**1**	**1**	0	**1**

128 no · 64 no · 32 no · 16 no · **8 yes** (5 left) · **4 yes** (1 left) · 2 no · **1 yes** (0 left). So **13 = 00001101**. Check: 8 + 4 + 1 = 13.

Task 2.1 · Decimal to binary

DO

NUMBER	128	64	32	16	8	4	2	1	CHECK
6									
20									
37									
64									
100									
255									

Reading binary, and a secret message

KEY SKILL

18 MIN

Task 2.2 · Binary to decimal

PROVE

Add up the value of every column that holds a 1.

BINARY	WORKING	ANSWER
00000111		
00010010		
00101000		
01000001		
10000001		
11111111		

Task 2.3 · Decode the secret message

PUZZLE

Letters are stored as numbers. Convert each byte to a decimal number, then use the table to find its letter.

BINARY	DECIMAL	LETTER
01000011		
01001111		
01000100		
01000101		

The word is _____

Letter codes

A 65 B 66 C 67 D 68
E 69 F 70 G 71 H 72
I 73 J 74 K 75 L 76
M 77 N 78 O 79 P 80
Q 81 R 82 S 83 T 84
U 85 V 86 W 87 X 88
Y 89 Z 90

Task 2.4 · Write your own initials

DO

Write your first and last initial as binary bytes.

Letter _____ = _____ = _____

Letter _____ = _____ = _____

LESSON 02 · COUNTING LIKE A COMPUTER

How big is a file, really?

STRETCH

16 MIN

bit	byte	KB	MB
one 0 or 1	8 bits · one letter	1,000 bytes	1,000 KB

Task 2.5 · Work it out

CALCULATE

#	QUESTION	WORKING	ANSWER
a	How many bits in 4 bytes?		
b	A text message is 160 letters. How many bytes?		
c	How many bytes in 3 KB?		
d	A song is 5 MB. How many KB?		
e	How many photos of 4 MB fit on a 64,000 MB phone?		

Task 2.6 · Why 255 keeps appearing

THINK

In Task 2.1 you wrote 255 as `11111111` — if you did it correctly, every single column was full.

a. What is the **largest** number you can store in 8 bits? _____

b. Screen colors are stored as three 8-bit numbers — one for red, one for green, one for blue. What is the brightest possible red?

Red = _____ Green = _____ Blue = _____

c. Why can a computer never store the number 300 in a single byte?

_____

_____

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

_____

_____

Teacher initials _____

Answer key

Counting Like a Computer

NOT FOR STUDENTS

ANSWERS

Task 2.1 · decimal to binary

answers	BINARY
6	00000110
20	00010100
37	00100101
64	01000000
100	01100100
255	11111111

Task 2.2 · binary to decimal

answers	DECIMAL
00000111	7 (4+2+1)
00010010	18 (16+2)
00101000	40 (32+8)
01000001	65 (64+1)
10000001	129 (128+1)
11111111	255

Task 2.3 · the secret message

decoded	ASCII
01000011	67 = C
01001111	79 = O
01000100	68 = D
01000101	69 = E

The word is CODE.

Task 2.5 · file sizes

a. 4 × 8 = **32 bits**

b. **160 bytes** (one byte per letter)

c. 3 × 1,000 = **3,000 bytes**

d. 5 × 1,000 = **5,000 KB**

e. 64,000 ÷ 4 = **16,000 photos**

Task 2.6 · the 255 question

a. 255. Eight columns holding 128+64+32+16+8+4+2+1. Note that 256 different values are possible, because zero counts — a common and forgivable confusion.

b. Red = 255, Green = 0, Blue = 0.

c. 300 is larger than 255, and 255 is the biggest number eight columns can hold. Storing it needs more bits. Accept any answer reaching “there are not enough bits”.

If a student asks about 1024

They are right, and they have been reading ahead. Strictly a kilobyte is 1,024 bytes, because computers work in powers of two. This lesson uses 1,000 to keep the arithmetic clean, which is also how storage manufacturers label their products. Both answers are defensible — tell them so.

LESSON 03 · TEACHER BRIEFING

Trace the Code

READ BEFORE THE BELL

2 MINUTES

Students work out what a program does by following it line by line on paper, filling in a trace table. This is exactly how a professional finds a bug, and it needs no computer at all.

You do not need to know this subject

The programs are five to eight lines long and every trace table is completed in the answer key, row by row. If you can follow a shopping list, you can check a trace table against the key.

Photocopy these pages only

Pages 17–19

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Working needs to be visible, so discourage erasing.

WHAT TO SAY TO START

> *"Programmers spend more time reading code than writing it. Today you are going to be the computer. You will follow a program one line at a time and write down what changes. No guessing — line by line."*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the worked example together	Do it on the board line by line — worth the time
14–26	Task 3.1 — the swap program	The result surprises them. Let it
26–38	Tasks 3.2 and 3.3 — loops and decisions	Key has every row filled in
38–48	Tasks 3.4 and 3.5	Harder. Pairs are fine here
48–55	Extension, collect sheets	Discussion prompt below

Questions students will ask

"Do I need to know Python?"
No. Everything you need is on the page. Say that out loud — several will assume they cannot start.

"What does the arrow mean in the table?"
It does not use arrows. Each row is one line of the program, and you write the new value of each variable after that line runs.

"What is a variable?"
A labeled box that holds one value at a time. Putting a new value in erases the old one. That erasing is what Task 3.1 is really about.

If they finish early

Task 3.5 asks students to find which input makes a program print a given answer — working backwards rather than forwards. After that, ask them to write a five-line program of their own and a completed trace table for a partner to check.

If the computers are working

Students can type each program in and run it to check their trace. Insist they complete the paper trace first, or the exercise loses its point entirely.

Curriculum focus · for the regular teacher's records

Tracing and desk-checking algorithms; variables, iteration and branching; predicting program output. Discussion prompt for the last five minutes: *"Why would a professional trace code on paper when the computer could just run it?"*

LESSON 03 · TRACE THE CODE

Be the computer

UNDERSTAND

14 MIN

NAME _____ CLASS _____ DATE _____

A trace table catches what your eyes miss

A **variable** is a labeled box holding one value. When you put a new value in, the old one is gone forever.

To trace a program you take one line at a time, in order, and write down what every box holds afterwards. Never skip ahead, and never guess the ending.

example.py — PYTHON

```
1 x = 2
2 y = 6
3 x = x + 4
4 y = y - x
5 print(x, y)
```

LINE	X	Y	OUTPUT
1	2	—	
2	2	6	
3	6	6	
4	6	0	
5	6	0	6 0

Line 3 replaces x with 2 + 4 = 6. Line 4 then uses the **new** x, not the old one. That is where nearly every mistake happens.

Task 3.1 · What does this really do?

TRACE

swap.py — PYTHON

```
1 a = 5
2 b = 3
3 a = a + b
4 b = a - b
5 a = a - b
6 print(a, b)
```

In one sentence, what has this program done to a and b?

LINE	A	B	OUTPUT
1			
2			
3			
4			
5			
6			

Loops and decisions

KEY SKILL

18 MIN

Task 3.2 · A loop that adds up

TRACE

total.py — PYTHON

```
total = 0
for n in range(1, 6):
    total = total + n
    print(total)
```

The loop runs once for n = 1, 2, 3, 4 and 5. It stops **before** 6.

The last number printed is _____

PASS	N	TOTAL	PRINTED
1			
2			
3			
4			
5			

Task 3.3 · One program, five results

TRACE

grade.py — PYTHON

```
if score >= 80:
    print("A")
elif score >= 60:
    print("B")
elif score >= 50:
    print("C")
else:
    print("N")
```

Python checks each condition in order and stops at the **first** one that is true.

SCORE	OUTPUT
92	
80	
79	
50	
49	

Which score is the hardest to get right, and why?

_____

Task 3.3b · One line changed

PREDICT

The program below is the same as grade.py, except the first two conditions have been swapped. Fill in the new results.

grade2.py — PYTHON

```
if score >= 60:
    print("B")
elif score >= 80:
    print("A")
else:
    print("N")
```

SCORE	OUTPUT
92	
65	
30	

Can this program **ever** print A? Explain.

_____

Harder traces

STRETCH

16 MIN

Task 3.4 · A loop inside a loop

PREDICT

nested.py — PYTHON

```
for i in range(1, 4):
    for j in range(1, 3):
        print(i, j)
```

The inside loop finishes completely before the outside loop moves on.

How many lines are printed in total? _____

Write every line, in order:

_____

_____

_____

_____

_____

_____

_____

Task 3.5 · Work backwards

CHALLENGE

count.py — PYTHON

```
x = 4
count = 0
while x > 0:
    x = x - 1
    count = count + 2
print(count)
```

a. What is printed? _____

b. How many times does the loop run? _____

c. What would x have to start as for the program to print 20? _____

d. What would happen if line 4 were deleted?

_____

_____

Task 3.6 · The last word

THINK

A computer could run all of these programs in less than a thousandth of a second. Write one sentence explaining why a professional would still trace one by hand.

_____

_____

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

_____

_____

Teacher initials _____

Answer key

Trace the Code

NOT FOR STUDENTS

ANSWERS

Task 3.1 · swap.py

LINE	A	B	OUT
1	5	—	
2	5	3	
3	8	3	
4	8	5	
5	3	5	
6	3	5	3 5

It swaps a and b without using a third variable.

Task 3.2 · total.py

PASS	N	TOTAL	PRINTED
1	1	1	1
2	2	3	3
3	3	6	6
4	4	10	10
5	5	15	15

Last number printed: **15**.

Task 3.3 · grade.py

92 → **A** · 80 → **A** (80 is not less than 80) · 79 → **B** · 50 → **C** · 49 → **N**

80 and 50 are the hard ones — the boundary values. `>=` includes the number itself. Boundaries are where real bugs live.

Task 3.4 · nested.py

output — PRINTED

```
1 1
1 2
2 1
2 2
3 1
3 2
```

Six lines — 3 outer passes × 2 inner passes.

Task 3.3b · grade2.py — the unreachable branch

92 → **B** · 65 → **B** · 30 → **N**

It can never print A. Any score of 80 or more is also 60 or more, so the first condition always catches it and Python stops there. The A branch is **unreachable code** — a real and surprisingly common bug, and the reason condition order matters.

Task 3.5 · count.py

a. 8. **b.** Four times — x goes 4, 3, 2, 1 and the loop stops when x reaches 0. **c.** x would start at 10, because count rises by 2 each pass and 10 × 2 = 20.

d. x would never change, so `x > 0` would stay true forever. The program would never stop — an **infinite loop**. Accept "it crashes" or "it freezes" as well.

The closing question

Running a program only tells you *what* it produced. Tracing tells you *where* it went wrong. A program that prints the wrong answer runs perfectly — the computer cannot help you find a logic error, so a human has to follow it by hand.

LESSON 04 · TEACHER BRIEFING

Flowchart Detective

READ BEFORE THE BELL

2 MINUTES

Students learn the four flowchart symbols, follow two flowcharts to work out what they do, find a genuine logic bug hiding in one of them, and then draw their own. Very visual, and easy to supervise.

You do not need to know this subject

Flowcharts are just boxes and arrows. The four shapes are explained on the first student page and the answer key covers every question. There is no code in this lesson at all.

Photocopy these pages only

Pages 22–24

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. A ruler is useful for Task 4.4 but not required.

WHAT TO SAY TO START

> *"Before anyone writes a program, they draw it. Today you are going to read three flowcharts, find a bug hidden in one of them that would cost a cafeteria real money, and then draw your own."*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the symbol panel together	Get them to name the four shapes aloud
12–24	Tasks 4.1 and 4.2 — following a flowchart	Answers are exact
24–36	Task 4.2 — the cafeteria bug	The bug is subtle. Hint is on this page if needed
36–48	Tasks 4.3 and 4.4 — loops, then draw your own	Rulers help but are not essential
48–55	Extension, collect sheets	Discussion prompt below

Questions students will ask

"Which shape do I use for that?"
Diamond for anything with a yes/no answer. Rectangle for an action. Slanted box for getting information in or showing it out. Rounded box only for start and end.

"Can a diamond have three arrows out?"
Not in this lesson. A decision has exactly two answers, yes and no. If it needs three, that is two decisions.

"I cannot find the bug in Task 4.2."
Hint to give: what happens if the food costs exactly the amount of money the student is holding?

If they finish early

Task 4.5 asks students to add error handling to their own flowchart — what happens if someone types a word instead of a number. Beyond that, ask them to flowchart logging in to a school computer, including what happens after three failed attempts.

If the computers are working

Any drawing tool works, and most students find drawing flowcharts on screen slower than on paper. The better on-screen task is to build one of the flowcharts as a working program in a block-based tool.

Curriculum focus · for the regular teacher's records

Representing algorithms as diagrams; branching and iteration; identifying and correcting logic errors. Discussion prompt for the last five minutes: *"The cafeteria bug loses about 20 cents at a time. Nobody notices. Why is that more dangerous than a program that crashes?"*

LESSON 04 · FLOWCHART DETECTIVE

The four shapes

UNDERSTAND

12 MIN

NAME	CLASS	DATE

START / END
Terminal
Where it begins and stops

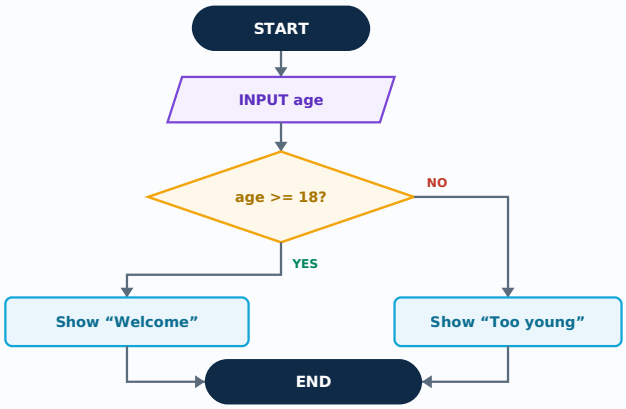
DO SOMETHING
Process
An action or a calculation

INPUT / OUTPUT
Data
Information in or out

YES / NO ?
Decision
Two ways out, always

Task 4.1 · Follow the flowchart

READ



Work through the diagram for each age and write what appears on screen.

AGE ENTERED	WHAT IS SHOWN
25	
18	
17	
0	

a. How many decisions does this flowchart contain? _____

b. Is it possible to reach END without seeing a message? _____

Find the bug that costs money

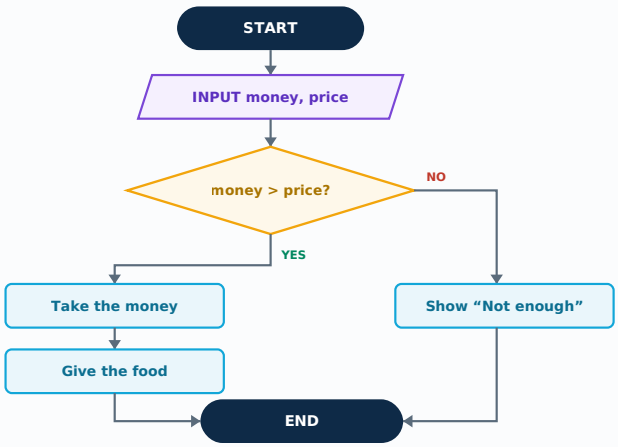
KEY SKILL

20 MIN

Task 4.2 · The school cafeteria

DETECTIVE

This flowchart runs the self-serve register at a school cafeteria. It works nearly all the time — but there is a logic error in it that quietly loses the cafeteria money.



First, follow it normally:

MONEY	PRICE	WHAT HAPPENS
$5.00	$3.50	
$2.00	$3.50	
$3.50	**$3.50**	

a. Which row above shows the problem?

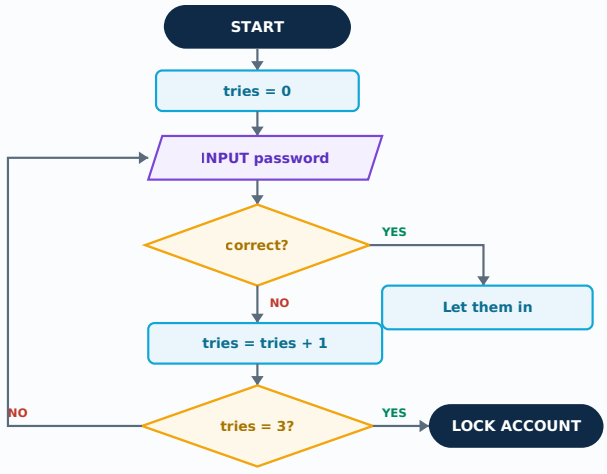
b. Describe what goes wrong in that case:

c. Write the one small change that fixes it:

Task 4.3 · The login screen

TRACE

This one controls a login screen. Notice the arrow that goes **backwards** — that is a loop.



a. How many times can someone get the password wrong before the account locks? _____

b. Which box makes the loop possible?

c. What would happen if the box `tries = tries + 1` were deleted?

d. Why is it safer to lock the account than to keep asking forever?

Draw your own

STRETCH

18 MIN

Task 4.4 · The bus fare machine

DRAW

Draw a flowchart for a bus fare machine that follows these rules. Use all four shapes.

- Ask the passenger for their age
- If they are under 5, the fare is free
- If they are 5 to 17, the fare is $2
- Otherwise the fare is $4
- Show the fare on the screen, then stop

Task 4.5 · What if they type nonsense?

EXTENSION

A real machine has to cope with a passenger who types “banana” instead of a number. Describe, in words, the extra decision you would add and where it would go.

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Flowchart Detective

Task 4.1 · following the flowchart

25 → **Welcome** · 18 → **Welcome** (18 is not less than 18) · 17 → **Too young** · 0 → **Too young**

a. One decision. **b.** No — every path shows a message before END.

Task 4.2 · the bug

The decision asks `money > price`. It should ask `money >= price`.

A student holding **exactly** $3.50 for a $3.50 sandwich is refused, because $3.50 is not *greater than* $3.50. The fix is to change the `>` to `>=`.

Task 4.2 · the three rows

$5.00 vs $3.50 → money taken, food given. Correct.

$2.00 vs $3.50 → "Not enough". Correct.

$3.50 vs $3.50 → **"Not enough" — and this is wrong.** Row c is the answer to part a.

Task 4.3 · the login loop

a. Three wrong attempts. Trace it: tries becomes 1, then 2, then 3, and only then does the decision send it to LOCK ACCOUNT.

b. The decision `tries = 3?` — its NO branch is the arrow that goes backwards to INPUT password. Accept an answer naming the backwards arrow itself.

c. tries would stay at 0 forever, so `tries = 3?` would never be true and the screen would ask for the password endlessly. An **infinite loop**, and the account would never lock.

d. Without a limit, an attacker can keep guessing until they get it right. Locking after three attempts makes guessing a password useless as a method.

Task 4.4 · grading the bus fare flowchart

Give credit for structure, not neatness. Look for:

- Rounded START and END boxes
- A slanted box for INPUT age
- **Two** diamonds, not one or three
- Every diamond has exactly two arrows out, labeled YES and NO
- A slanted box showing the fare
- All three paths reach END

The usual order is: *age < 5?* → yes, free. No → *age < 18?* → yes, $2. No → $4. Any equivalent ordering that produces the same three outcomes is correct.

Task 4.5 · model answer

Add a decision immediately after INPUT age asking *"is this a number?"*. If NO, show an error and loop back to the input box. If YES, carry on to the fare decisions. Accept any answer that puts the check **before** the fare is calculated — that placement is the whole idea.

LESSON 05 · TEACHER BRIEFING

How Attacks Actually Work

READ BEFORE THE BELL

2 MINUTES

Students rank passwords, dissect three suspicious emails, and work out how an attacker gets into a school without writing a line of code. The most discussion-friendly lesson here.

You do not need to know this subject

Nothing technical is required. Every answer is a judgment about whether something looks suspicious, and the key lists every warning sign to look for in each email.

Photocopy these pages only

Pages 27–29

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Do not let students write down any real password.

WHAT TO SAY TO START

> *"Almost nobody gets hacked by a genius typing furiously in a dark room. They get hacked because they used the same password twice, or because somebody polite asked them for it. Today you find out how it actually happens."*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read 'three ways in' together	Ask who has reused a password. Not which
12–24	Task 5.1 — ranking passwords	Expect argument. Argument is good here
24–38	Task 5.2 — the three emails	Key lists every warning sign
38–50	Task 5.3 — social engineering	Best run as a class discussion if the room allows
50–60	Task 5.4, collect sheets	Discussion prompt below

Questions students will ask

"Is my password safe?"
Never let a student say or write a real password. Redirect it: is a password *like* that one safe?

"What if I have already been hacked?"
Change the password, turn on two-step verification, tell a parent or school IT.

"Can you teach us to hack?"
This lesson is recognizing an attack, which is the same knowledge from the other side. Attacking a real system without permission is a crime.

If they finish early

Task 5.3 asks for a school password policy. Strong students can then argue why forcing a password change every 30 days makes a school *less* secure. Answer in the key.

If the computers are working

Students can test password strength on a reputable strength meter. Insist they use invented passwords, never a real one, and never one they intend to use.

Curriculum focus · for the regular teacher's records

Cyber security threats and safeguards; social engineering and phishing. Discussion prompt for the last five minutes: *"The strongest password is useless typed into the wrong site. So what actually protects you?"*

LESSON 05 · HOW ATTACKS ACTUALLY WORK

Three ways in

UNDERSTAND

14 MIN

NAME	CLASS	DATE

1 · Guess it

A computer can try billions of passwords a second. Short passwords and real words fall in moments.

2 · Trick you

Far easier than guessing. A convincing email or a polite phone call and you hand it over yourself.

3 · Find it lying around

A sticky note, an unlocked screen, a password reused on a site that has already been breached.

Task 5.1 · Rank these passwords

JUDGE

Rank them 1 (safest) to 8 (worst). Then write what is wrong with the four weakest.

PASSWORD	RANK	WHAT IS WRONG WITH IT
Password1!		
correct-horse-battery		
Jayden2009		
xk4%Lm9		
123456		
MyDogRustyIsGreat!		
qwertyuiop		
P@ssw0rd		

Two of these passwords are long but made of ordinary words. Are they weak?

_____

Rule for this lesson

Never write a real password on this sheet and never say one out loud. Everything here is invented.

LESSON 05 · HOW ATTACKS ACTUALLY WORK

Spot the fake

KEY SKILL

18 MIN

All three emails below arrived this morning. For each one, circle the parts that give it away and list them underneath.

EMAIL 1

From: service@paypal-secure.com
Subject: Your account has been suspended

Dear Valued Customer,
We have detect unusual activity on you account. Your account will be permanently closed within **24 hours** unless you verify your identity immediately.

Click here to restore access

Thank you, Security Team

Warning signs:

EMAIL 2

From: principal@lincolnhigh-admin.net
Subject: URGENT - need this before 3pm

Hi,
I am in a meeting and cannot take calls. I need you to buy four $100 gift cards for a staff award and send me the codes. I will reimburse you tomorrow.

Please keep this between us until the assembly.

Sent from my iPhone

Warning signs:

EMAIL 3

From: noreply@yourschool.edu
Subject: Your schedule for the fall semester

Hi students,
Your fall semester schedule is now available in the student portal. Log in the usual way from the school website.

No password is required in this email.

Student Services

This one is different. What is different about it?

The five things to check, every time

1. **The sender address**, not the sender name. Look character by character.
2. **Urgency.** Real organizations do not give you 24 hours.
3. **Secrecy.** “Keep this between us” is the single biggest warning sign there is.
4. **The ask.** Gift cards, passwords, bank details — never by email.
5. **Language.** Odd grammar, missing name, generic greeting.

Which of the five is the strongest signal on its own? _____

LESSON 05 · HOW ATTACKS ACTUALLY WORK

The polite attack

STRETCH

18 MIN

Task 5.2 · What would you do?

DECIDE

None of these involve any technical skill at all. For each, write what you would actually do.

#	SITUATION	WHAT YOU WOULD DO
a	Someone in a high-visibility vest asks you to hold the gate open because their pass is not working.	
b	A caller says they are from IT and needs your password to fix your account before it locks.	
c	You find a USB drive in the parking lot labeled *Grade 10 exam*.	
d	A friend asks to borrow your login “just for one class”.	

Task 5.3 · Write the school password rule

EXTENSION

Write three rules for student passwords at your school. Each one must be a rule people will actually follow — a rule everybody ignores protects nobody.

1. _____
2. _____
3. _____

Task 5.4 · The last word

THINK

Some schools force everyone to change their password every 30 days. Explain in one sentence why that might make the school **less** safe, not more.

_____

_____

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

_____

_____

Teacher initials _____

Answer key

How Attacks Actually Work

NOT FOR STUDENTS

ANSWERS

Task 5.1 · password ranking

The exact order is arguable; the reasoning is what earns the credit. A defensible ranking:

RANK	PASSWORD	WHY
1	`correct-horse-battery`	Long. Length beats complexity
2	`MyDogRustyIsGreat!`	Long, but the dog is guessable if known
3	`xk4%Lm9`	Random but far too short
4	`Jayden2009`	Name plus birth year — both public
5	`P@ssw0rd`	Every cracking tool knows this substitution
6	`Password1!`	The most common ‘strong’ password there is
7	`qwertyuiop`	A straight line on the keyboard
8	`123456`	Consistently the most used password in the world

The ordinary-words question: no, they are not weak. Four random common words are far harder to crack than seven scrambled characters, because **length matters more than strangeness**. That surprises most classes and is worth saying explicitly.

Email 1

Sender address uses a capital **I** in place of the **l** in PayPal · “we have detect” and “you account” are broken grammar · generic “Valued Customer” · 24-hour deadline · a link instead of asking you to log in normally.

Email 2

Domain is `.net` and not the school’s real address · urgency · **secrecy** · gift cards, which are the single most common request in this scam · cannot be phoned to check · no name used.

Email 3

This one is **legitimate**. It asks for nothing, contains no link, tells you to log in the usual way, and creates no urgency. Students who flag it as fake have learned suspicion but not judgment — worth discussing.

Strongest single signal

Secrecy. A genuine organization never asks you to keep a request from your colleagues, your parents or your school. Urgency is close behind.

Task 5.2 · model responses

a. Do not hold the gate. Direct them to the office. Not your job to verify anyone, and that is precisely why it works.

b. Hang up and call the school on a number you looked up yourself. Real IT staff never need your password.

c. Hand it to a teacher unopened. Plugging in a found USB drive is one of the oldest and most effective attacks there is.

d. No. Anything they do is recorded as you — that is the part students usually have not considered.

The 30-day password question

Forcing frequent changes pushes people toward weak, predictable patterns — `Summer1, Summer2, Summer3` — and toward writing them down. Most security guidance now recommends one long, strong password changed only when there is reason to think it has leaked. Accept any answer reaching “people cope by choosing worse passwords”.

LESSON 06 · TEACHER BRIEFING

Sending a Message Across the World

READ BEFORE THE BELL

2 MINUTES

Students break a message into numbered packets, reassemble a jumbled set, find the cheapest route across a network map, and work out what happens when a packet is lost.

You do not need to know this subject

A puzzle lesson. The packet tasks are self-checking — the message either reads or it does not — and the routing is simple addition. Every answer is in the key.

Photocopy these pages only

Pages 32–34

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Nothing else.

WHAT TO SAY TO START

> *"When you send a message, it does not travel in one piece. It is chopped up, the pieces take different routes, and they often arrive in the wrong order. Today you find out how they get put back together."*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the packet panel together	The postcard analogy does most of the work
14–26	Tasks 6.1 and 6.2 — splitting and reassembling	Self-checking
26–40	Task 6.3 — the network map	Simple addition. Only one route is cheapest
40–50	Task 6.4 — lost packets	Key has model answers
50–60	Extension, collect sheets	Discussion prompt below

Questions students will ask

"Why not send the whole message at once?"
Lose a big message and you resend all of it. Lose one packet and you resend one packet. Small pieces also let many people share one cable.

"How does it know which route to take?"
Each router only knows which neighbor is closest. No machine knows the whole path — which is why the internet survives outages.

"Is there really more than one route?"
Yes. Packets from one message can take different routes, which is why they arrive out of order.

If they finish early

Task 6.4 covers lost packets and why video calls go blurry while texts always arrive. Strong students can find the second cheapest route and say when it would be used.

If the computers are working

Students can run `tracert` (Windows) or `traceroute` (Mac) to a well-known website and count the real hops. It usually surprises them how many there are.

Curriculum focus · for the regular teacher's records

Transmitting data across networks; packets, addressing and routing; reliability. Discussion prompt for the last five minutes: *"No computer knows the whole route your message takes. Why does that make the internet harder to break?"*

LESSON 06 · SENDING A MESSAGE ACROSS THE WORLD

Chopping a message into pieces

UNDERSTAND

14 MIN

NAME	CLASS	DATE

Nothing travels in one piece

Imagine mailing a long letter as forty numbered postcards. Each one carries the destination address, its own number, and a small piece of the message.

They go through different sorting offices, arrive in the wrong order, and the person at the other end uses the numbers to rebuild the letter. That is a **packet**, and it is exactly how the internet works.

Task 6.1 · Break it up

DO

Split this message into packets of **four characters each**, counting spaces. Number them in order.

message — 19 CHARACTERS

MEET ME AT THE GATE

PACKET #	CONTENTS	PACKET #	CONTENTS
1		4	
2		5	
3		6	

How many packets did you need in total? _____

Task 6.2 · They arrived in the wrong order

REBUILD

These six packets arrived jumbled. Put them back together using the numbers.

ARRIVED	PACKET #	CONTENTS
1st	4	E CO
2nd	1	THE
3rd	6	WN
4th	2	SERV
5th	5	ME D
6th	3	ER I

The message is _____

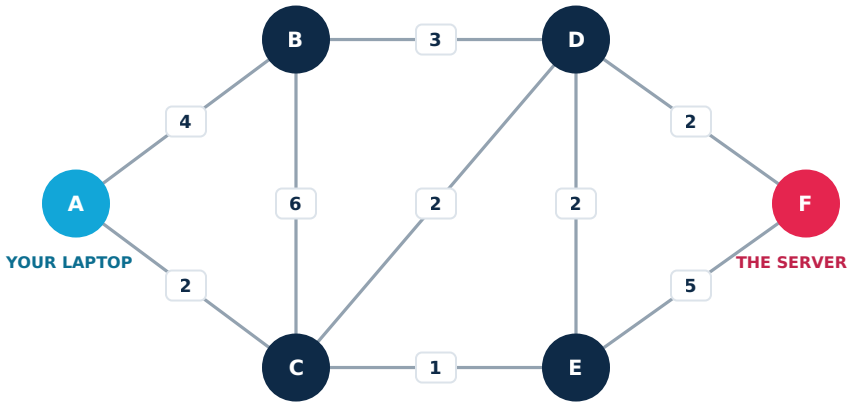
Finding the way

KEY SKILL

18 MIN

Every link has a cost

Each line below is a cable between two routers. The number is how long that hop takes. Your laptop is **A**. The server you are trying to reach is **F**.



Task 6.3 · Find the cheapest route

CALCULATE

Add the numbers along each route. The cheapest total wins.

ROUTE	WORKING	TOTAL
A → B → D → F		
A → C → E → F		
A → C → D → F		
A → B → C → E → F		

a. The cheapest route is _____ with a total of _____

b. The cable between C and E breaks. Which route would the network use instead, and what does it cost now? _____

When things go missing

STRETCH · 16 MIN

The receiving computer checks the numbers

If packets 1, 2, 4 and 5 arrive but 3 does not, the receiver knows immediately that something is missing — and asks for that one packet again. It does not ask for the whole message.

Task 6.4 · Missing pieces

THINK

a. A message was split into 8 packets. Packets 1, 2, 3, 5, 6, 7 and 8 arrive. What does the receiving computer do?

b. Why is it better to resend one packet than the whole message?

c. A text message always arrives perfectly, but a video call goes blurry and freezes. Both use the same internet. Why the difference?

Task 6.5 · The order of events

EXTENSION

Number these steps 1 to 6 to show what happens when you open a web page.

ORDER	STEP
	The server sends the page back, split into packets
	Your browser reassembles the packets and draws the page
	You type an address and press Enter
	Routers pass the request along, hop by hop
	Your computer looks up the address to find the server’s number
	Your computer sends a request packet toward that number

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Sending a Message Across the World

NOT FOR STUDENTS

ANSWERS

Task 6.1 · splitting the message

MEET ME AT THE GATE is 19 characters including spaces.

packets	FOUR AT A TIME
1 = MEET	
2 = ME (space, M, E, space)	
3 = AT T	
4 = HE G	
5 = ATE	

5 packets. The last one is short, which is normal — real packets are rarely all full.

Task 6.2 · reassembled

in number order	REBUILT
1 THE · 2 SERV · 3 ER I	
4 E CO · 5 ME D · 6 WN	

Joined in number order it reads **THE SERVER IS COME DOWN**.

Students who reassemble by **arrival order** instead of packet number get nonsense, which is exactly the point of the task. If a student notices the grammar is odd, they have spotted that a corrupted packet still reassembles — the checksum problem. Give them credit for it.

Task 6.3 · route costs

totals	ADDITION
A-B-D-F = 4+3+2 = 9	
A-C-E-F = 2+1+5 = 8	
A-C-D-F = 2+2+2 = 6	
A-B-C-E-F = 4+6+1+5 = 16	

a. Cheapest is **A → C → D → F**, total **6**.

b. Losing C–E does not matter — the cheapest route never used it. The network still sends via A→C→D→F at a cost of 6. Students who spot that get full credit and have understood redundancy.

Task 6.4 · model answers

a. It notices packet 4 is missing and asks the sender for that one packet again.

b. Far less data to resend, and far less time wasted. On a long download you would otherwise start again from the beginning.

c. A text message waits for every packet and resends anything missing, because a late message is fine. A video call cannot wait — a frame that arrives late is useless, so it drops it and carries on. That is why it goes blurry instead of pausing.

Task 6.5 · correct order

1. You type an address and press Enter · **2.** Your computer looks up the address · **3.** It sends a request packet · **4.** Routers pass it along hop by hop · **5.** The server sends the page back in packets · **6.** Your browser reassembles and draws it.

LESSON 07 · TEACHER BRIEFING

Sorting and Searching by Hand

READ BEFORE THE BELL

2 MINUTES

Students perform a bubble sort by hand and count every swap, then race a linear search against a binary search on the same list. The difference is dramatic and they discover it themselves rather than being told.

You do not need to know this subject

This is counting and comparing, nothing more. Every pass of the sort is written out in the answer key so you can check a student's working row by row.

Photocopy these pages only

Pages 37–39

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Erasing makes the working impossible to check, so ask them to cross out instead.

WHAT TO SAY TO START

> *"Your phone can search a list of a billion things in under a second. It does not look at them all — that would take a week. Today you find out what it actually does, and you do it yourself with a pen."*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the bubble sort worked example	Do the first pass on the board with them
14–30	Task 7.1 — sorting by hand	Slow and fiddly by design. Key has every pass
30–42	Task 7.2 — linear vs binary search	The step counts are the payoff
42–52	Task 7.3 — the doubling table	This is where it clicks for most students
52–60	Extension, collect sheets	Discussion prompt below

Questions students will ask

"Do I really have to write out every pass?"
Yes. The whole point is feeling how much work it is. Students who shortcut it miss the lesson entirely.

"Why is the list already sorted for the search task?"
Because binary search only works on a sorted list. That is the trade-off the lesson is building toward — sorting costs effort but makes every future search cheap.

"Is bubble sort what computers actually use?"
No, and that is worth saying. It is the easiest to do by hand, which is why it is taught. Real systems use faster methods built on the same ideas.

If they finish early

Task 7.4 extends the doubling table to a million items. Strong students can then work out how many people you could find in a city phone book with 21 questions, and explain why doubling the list only adds one step.

If the computers are working

Students can time a search on a large spreadsheet, sorted and unsorted, using the find function. The difference is measurable even on a few thousand rows.

Curriculum focus · for the regular teacher's records

Algorithm efficiency; comparing algorithms that solve the same problem; the relationship between input size and number of steps. Discussion prompt for the last five minutes: *"Sorting a list takes effort. Why is it worth doing before you search it even once?"*

Sorting the slow way

UNDERSTAND

18 MIN

NAME _____ CLASS _____ DATE _____

Bubble sort: compare two neighbors, swap if they are the wrong way round

Start at the left. Compare the first two numbers. If the left one is bigger, swap them. Move one place right and repeat, all the way to the end. That is one **pass**. Then start again from the left.

You stop when a whole pass produces no swaps at all.

Worked example · first pass on 5 1 4 2

COMPARE	LIST	SWAP?
5 and 1	5 1 4 2	Yes → 1 5 4 2
5 and 4	1 5 4 2	Yes → 1 4 5 2
5 and 2	1 4 5 2	Yes → 1 4 2 5

After one pass the largest number has **bubbled** to the end. That always happens, and it is where the name comes from.

Task 7.1 · Sort this list completely

DO

Starting list: `8 3 7 1 5 2`. Write the list at the end of each pass, and count the swaps.

PASS	LIST AFTER THIS PASS	SWAPS
1		
2		
3		
4		
5		

a. How many passes until no swaps happen? _____ **b.** Total swaps altogether? _____

LESSON 07 · SORTING AND SEARCHING BY HAND

Two ways to find something

KEY SKILL

16 MIN

The sorted list

fifteen names — ALREADY IN ORDER

1. ADAMS 2. BROWN 3. CHEN 4. DIAZ
5. EVANS 6. FOX 7. GRANT 8. HUANG
9. IRWIN 10. JONES 11. KHAN 12. LEE
13. MURPHY 14. NGUYEN 15. OWENS

Method A · Linear search

Start at number 1 and check every name in turn until you find it. Simple, and it works on any list.

Method B · Binary search

Jump to the middle. If your name comes earlier in the alphabet, throw away the whole second half. Repeat with what is left. Only works if the list is sorted.

Task 7.2 · Count the steps

RACE

One step means one name checked. Do it properly — do not estimate.

FIND THIS NAME	LINEAR STEPS	BINARY STEPS	WHICH WON?
ADAMS			
OWENS			
KHAN			
CHEN			

Which name is the **only** one where linear search wins, and why? _____

Task 7.2b · Show your halving

PROVE

Write out the binary search for **GRANT**. At each step, say which name you checked and which half you threw away.

STEP	NAME CHECKED	HALF KEPT
1		
2		
3		
4		

Roughly how many names were eliminated by the very first check? _____

Why it matters at scale

STRETCH · 16 MIN

Task 7.3 · The doubling table

CALCULATE

Each time you double the list, binary search needs only **one** extra step. Fill in the gaps.

LIST SIZE	LINEAR · WORST CASE	BINARY · WORST CASE
15 names	15	4
30 names		
60 names		
1,000 names		10
1,000,000 names		20

a. Going from 1,000 to 1,000,000 names multiplies the list by a thousand. How many extra steps does binary search need? _____

Task 7.4 · The catch

THINK

a. Binary search is enormously faster. Why can you not always use it?

b. Sorting a list takes real effort. Explain why it is still worth sorting a list you plan to search many times.

c. You have a list of 1,000 names and you need to search it exactly **once**. Should you sort it first? Explain.

Task 7.5 · The last word

THINK

In one sentence: why is bubble sort taught in schools even though no professional would ever use it?

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Sorting and Searching by Hand

NOT FOR STUDENTS

ANSWERS

Task 7.1 · bubble sort, every pass

PASS	LIST AFTER THIS PASS	SWAPS
1	3 7 1 5 2 8	5
2	3 1 5 2 7 8	3
3	1 3 2 5 7 8	2
4	1 2 3 5 7 8	1
5	1 2 3 5 7 8	0 · stop

a. Five passes — the fifth produces no swaps, which is how you know it is finished. **b.** Eleven swaps in total. A student who stops after pass 4 has the right list but has not proved it is sorted; give partial credit and explain the difference.

Task 7.2 · step counts

NAME	LINEAR	BINARY
ADAMS	1	4
OWENS	15	4
KHAN	11	3
CHEN	3	4

ADAMS is the only clear win for linear search, because it is the very first name. CHEN is also close. Accept small variations in the binary counts — the halving point can be taken slightly differently — as long as the working is shown.

Task 7.3 · the doubling table

SIZE	LINEAR	BINARY
15	15	4
30	30	5
60	60	6
1,000	1,000	10
1,000,000	1,000,000	20

a. Ten extra steps. The list grew a thousandfold; the work grew by ten.

Task 7.2b · binary search for GRANT, step by step

1. Check 8, HUANG. GRANT comes earlier → keep 1–7. **2.** Check 4, DIAZ. GRANT comes later → keep 5–7. **3.** Check 6, FOX. GRANT comes later → keep 7 only. **4.** Check 7, GRANT. Found.

The first check eliminated **eight** names — over half the list — in a single step. That is the whole idea. Accept slightly different halving points provided the working is shown.

Task 7.4 · model answers

a. The list has to be sorted first. On an unsorted list, throwing away half is meaningless — the name could be anywhere.

b. Sorting is a one-off cost. Every search afterwards is cheap, so the effort is repaid many times over.

c. No. Sorting 1,000 names costs far more than one linear search through them. Sort only when you will search repeatedly — the best question on the page, because it weighs cost against benefit.

Task 7.5 · the closing question

Bubble sort is taught because it can be done by hand and watched happening — its logic is visible in a way faster methods are not.

LESSON 08 · TEACHER BRIEFING

Bug Hunt

READ BEFORE THE BELL

2 MINUTES

Students learn the three kinds of program bug, classify six real error messages, then find and fix six broken programs. The last one runs perfectly and still gives the wrong answer, which is the most important bug of all.

You do not need to know this subject

Every bug is a single line, and the answer key names the line, the bug type and the fix for each one. You are checking answers against a key, not diagnosing code.

Photocopy these pages only

Pages 42–44

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Pairs are fine and productive for this lesson.

WHAT TO SAY TO START

> *"Every program you have ever used had bugs in it. Professional programmers spend more time fixing than writing. Today you get six broken programs and you have to work out what is wrong with each one."*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the three bug types together	The distinction matters more than the code
14–24	Task 8.1 — classify the error messages	Six answers, all in the key
24–42	Task 8.2 — the six broken programs	One bug each. Pairs work well here
42–52	Task 8.3 — the program that lies	Hardest and most important task on the sheet
52–55	Collect sheets	Discussion prompt below

Questions students will ask

"The program looks fine to me."
For Task 8.3 that is the correct reaction — it does look fine. Tell them to work out the answer by hand first, then compare.

"Do I need to know Python?"
No. Every bug is visible from the plain-English comment beside the code. Say this before they start or half the class will not begin.

"Is a spelling mistake a bug?"
In code, yes — a misspelled variable name is one of the commonest bugs there is. In a message shown to the user, it is a bug too, just a cosmetic one.

If they finish early

Task 8.4 asks students to write a test plan — what values would you feed a program to prove it works. Strong students should be pushed toward boundary values: zero, negative numbers, and the exact number where behavior changes.

If the computers are working

Students can type each broken program in and read the real error message, which is far more convincing than the printed one. Ask them to predict the message before they run it.

Curriculum focus · for the regular teacher's records

Testing and debugging; syntax, runtime and logic errors; developing test data. Discussion prompt for the last five minutes: *"A program that crashes tells you something is wrong. A program that quietly gives the wrong answer does not. Which would you rather have?"*

LESSON 08 · BUG HUNT

Three kinds of broken

UNDERSTAND

14 MIN

NAME	CLASS	DATE

1 · Syntax error

The code breaks the rules of the language. It will not run at all.

`print("hello"`

Missing bracket. Nothing happens.

2 · Runtime error

The code is legal, but something impossible happens while it runs.

`print(10 / 0)`

Starts, then crashes.

3 · Logic error

It runs perfectly and gives the wrong answer. The dangerous one.

`area = w + h`

No error. Just wrong.

Worked example · the same program, three ways to break it

syntax — PYTHON

```
if score >= 50
    print("PASS")
```

Missing colon. Will not run at all.

runtime — PYTHON

```
score = int("banana")
if score >= 50:
```

Starts, then crashes on line 1.

logic — PYTHON

```
if score >= 5:
    print("PASS")
```

Runs perfectly. Passes everyone with 5 or more.

Task 8.1 · Name that bug

CLASSIFY

Write **S** for syntax, **R** for runtime or **L** for logic beside each one.

#	WHAT HAPPENED	S / R / L
a	The program will not start. The editor highlights a missing colon.	
b	It calculates the average of 5 numbers by dividing by 4.	
c	It crashes when the user types a word instead of a number.	
d	A variable name is spelled `totl` in one place and `total` in another.	
e	It prints every student twice.	
f	It stops when it tries to open a file that has been deleted.	

LESSON 08 · BUG HUNT

Six broken programs

KEY SKILL

18 MIN

Each program has **exactly one** bug. Find it, name the type, and write the fix.

bug 1 — PYTHON

```
# Print a greeting five times
for i in range(5)
    print("Hello")
```

Type: _____ The fix:

bug 2 — PYTHON

```
# Add up three scores
m1 = 10
m2 = 20
m3 = 30
total = m1 + m2 + m4
print(total)
```

Type: _____ The fix:

bug 3 — PYTHON

```
# Ask for a number and double it
n = input("Number: ")
print(n * 2)

# User types 5, program prints 55
```

Type: _____ The fix:

bug 4 — PYTHON

```
# Count down from 5 to 1
count = 5
while count > 0:
    print(count)

# runs forever
```

Type: _____ The fix:

bug 5 — PYTHON

```
# Pass score is 50 or above
if score > 50:
    print("PASS")
else:
    print("FAIL")
```

Type: _____ The fix:

bug 6 — PYTHON

```
# Print the numbers 1 to 10
for i in range(1, 10):
    print(i)

# only prints up to 9
```

Type: _____ The fix:

The program that lies

STRETCH

16 MIN

Task 8.3 · It runs. It is wrong.

HARDEST

This program calculates a class average. It never crashes. It always produces a number. That number is wrong.

average.py — PYTHON

```
scores = [60, 70, 80, 90]
total = 0

for m in scores:
    total = total + m

average = total / 3
print("Average:", average)
```

a. Work out the correct average by hand:

_____

b. What does the program print instead?

_____

c. Which line is wrong? _____

d. Write the corrected line:

_____

e. Why is this kind of bug more dangerous than a crash?

_____

_____

Task 8.4 · Write a test plan

EXTENSION

A program takes a student's score and prints PASS if it is 50 or more. List five values you would test it with, and what each one proves.

TEST VALUE	WHAT IT PROVES

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

_____

_____

Teacher initials _____

Answer key

Bug Hunt

NOT FOR STUDENTS

ANSWERS

Task 8.1 · classification

a. S — missing colon, will not run
b. L — runs fine, answer wrong
c. R — legal code, crashes on bad input
d. S or **R** — accept either, with reasoning
e. L — no error, wrong output
f. R — crashes while running

About question d

A misspelled variable is often reported as a runtime error rather than a syntax one, because the program starts before it discovers the name does not exist. Both answers are defensible; the reasoning is what you are grading.

Task 8.2 · the six bugs

#	TYPE	THE BUG	THE FIX
1	Syntax	No colon after the for line	`for i in range(5):`
2	Syntax	m4 does not exist; should be m3	`total = m1 + m2 + m3`
3	Logic	input gives text, so “5” doubled is “55”	`n = int(input("Number: "))`
4	Logic	count never changes, so the loop never ends	`count = count - 1` inside the loop
5	Logic	A score of exactly 50 fails	`if score >= 50:`
6	Logic	range stops before 10	`for i in range(1, 11):`

Bugs 5 and 6 are both **off-by-one** errors, the single most common logic bug in programming. Worth naming out loud — students remember the phrase.

Task 8.3 · the program that lies

a. (60+70+80+90) ÷ 4 = 300 ÷ 4 = **75**

b. 300 ÷ 3 = **100**

c. The line `average = total / 3`

d. `average = total / len(scores)` — accept `total / 4`, but point out that it breaks the moment a fifth score is added. That distinction is worth credit on its own.

e. A crash tells you immediately that something is wrong. This program reports a confident, plausible number — 100 is a believable average — so nobody investigates. Wrong answers that look right can go unnoticed for years.

Task 8.4 · a good test plan

VALUE	WHAT IT PROVES
49	Just below the boundary — should FAIL
50	Exactly on the boundary — should PASS
51	Just above the boundary — should PASS
0	The lowest sensible value
100 or -5 or “banana”	The top of the range, or invalid input it must survive

Full credit requires at least one **boundary** value (49, 50, 51). Testing only 20 and 80 proves very little, because bugs almost never live in the middle of a range.

LESSON 09 · TEACHER BRIEFING

What AI Can and Cannot Do

READ BEFORE THE BELL

2 MINUTES

Students find out what an AI chatbot is actually doing when it answers, sort tasks into safe and risky uses, examine a confident paragraph that cannot be verified, and write their own rules for using AI in schoolwork.

You do not need to know this subject

This is a discussion and judgment lesson. There is no technology involved and no correct opinion — the answer key gives you the reasoning to look for rather than a single right answer.

Photocopy these pages only

Pages 47–49

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. No devices needed, even if the lesson is about them.

WHAT TO SAY TO START

> *"An AI chatbot does not know anything. It predicts what words usually come next. That sounds like a small difference. Today you will find out why it changes everything about how you should use it."*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read 'what it is actually doing'	The autocomplete comparison does the work
14–26	Task 9.1 — sorting tasks by risk	Expect disagreement. Ask for reasons, not votes
26–40	Task 9.2 — the confident paragraph	The point is that none of it can be checked
40–50	Task 9.3 — is it cheating?	The most valuable discussion in the pack
50–60	Task 9.4, collect sheets	Discussion prompt below

Questions students will ask

"Is using AI cheating?"
The honest answer is that it depends entirely on what was asked of you. Task 9.3 is designed to make students argue that out rather than be told.

"Does AI know it is wrong?"
No. It has no way to tell the difference between a true sentence and a plausible one. That is the single most important idea in the lesson.

"Do teachers use AI?"
Many do, for planning and drafting. Being straightforward about that earns far more credibility than pretending otherwise.

If they finish early

Task 9.4 asks students to write five rules for AI use at school. Strong students should then be asked which of their own rules would be impossible to enforce — that is a harder and more honest question than writing the rules.

If the computers are working

If AI tools are permitted at your school, students can ask a chatbot the same factual question three times and compare the answers. Inconsistency between the three is the clearest possible demonstration of the point.

Curriculum focus · for the regular teacher's records

Impacts of digital systems on society; evaluating information sources; ethical use of technology; data and bias. Discussion prompt for the last five minutes: *"If an AI is confident and wrong, and you cannot tell the difference, whose fault is the mistake?"*

LESSON 09 · WHAT AI CAN AND CANNOT DO

It is predicting, not knowing

UNDERSTAND

16 MIN

NAME	CLASS	DATE

The most useful thing to understand about AI chatbots

When your phone suggests the next word in a message, it is not reading your mind. It has seen millions of messages and it knows which word usually follows.

An AI chatbot is doing the same thing, on an enormous scale. It produces the words that most plausibly come next. It is extraordinarily good at this — but **plausible is not the same as true**, and the system has no way of telling the two apart.

Task 9.1 · Sort these by risk

JUDGE

Check one column for each task, then write your reason.

TASK	SAFE	RISKY	WHY
Suggesting ideas for a story			
Telling you the date of a historical event			
Rewriting your paragraph more clearly			
Giving medical advice about a symptom			
Explaining a concept you did not understand in class			
Writing your assignment for you			
Summarizing a page you have already read			
Naming the sources for a research task			

Look at everything you marked **safe**. What do those tasks have in common?

_____

Confident and unverifiable

KEY SKILL

18 MIN

An AI was asked to write about an American inventor

Read this carefully. It is well written, specific and completely confident.

> *Margaret Ellwood Harrow was born in Peoria, Illinois in 1897 and is widely credited with inventing the rotary seed drill. After studying at the Peoria School of Mines, she filed her first patent in 1923 and by 1931 her design was in use on more than 4,000 farms across Illinois and Iowa. The Harrow Drill Company employed 212 people at its peak. She was awarded the Dearborn Medal in 1948, and a street in East Peoria still bears her name.*

Every sentence sounds like a fact. Every sentence has a specific detail attached. That is exactly what makes it convincing.

Task 9.2 · What would you have to check?

INVESTIGATE

You are not being asked whether this is true. You are being asked what you would have to **check**, and how.

#	A CLAIM THAT NEEDS CHECKING	HOW WOULD YOU CHECK IT?
1		
2		
3		
4		
5		

Which single detail in the paragraph would be **fastest** to check, and why?

_____

The point of this task

A wrong answer that sounds uncertain is easy to catch. A wrong answer delivered with names, dates and numbers is not. Confidence is not evidence.

LESSON 09 · WHAT AI CAN AND CANNOT DO

Whose work is it?

STRETCH

18 MIN

Task 9.3 · Cheating, or not?

DECIDE

Mark each one **OK**, **NOT OK** or **DEPENDS**, then give your reason. There is genuine disagreement about some of these.

#	WHAT THE STUDENT DID	VERDICT	REASON
a	Asked AI to explain a topic, then wrote the essay themselves		
b	Asked AI to write the essay, then changed a few words		
c	Used AI to check their spelling and grammar		
d	Asked AI for five ideas, picked one, researched it properly		
e	Used AI to summarize a book they were meant to read		

Task 9.4 · Write the rules

EXTENSION

Write three rules for using AI on schoolwork that you think are **fair** — rules you would accept even when they worked against you.

1. _____
2. _____
3. _____

Now the hard part: which of your three rules would be **impossible for a teacher to enforce**, and does that make it a bad rule?

_____

_____

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

_____

_____

Teacher initials _____

Answer key

What AI Can and Cannot Do

NOT FOR STUDENTS

ANSWERS

Task 9.1 · what to look for

Generally safe — tasks where **you** already know the material and can spot a mistake: generating ideas, rewording your own writing, summarizing something you have read, explaining a concept you can then check against your notes.

Generally risky

Tasks where you **cannot tell** whether the answer is right: historical dates, medical advice, naming sources, and anything you are supposed to have produced yourself.

The common thread

AI is safest where you are able to check it. That is the sentence to draw out of the class.

Task 9.2 · important

Margaret Ellwood Harrow does not exist. The paragraph is entirely invented — the person, the company, the patent, the medal and the street. Tell the class this at the end, not the beginning. The uncomfortable moment when they realize how convincing it was is the lesson.

Task 9.2 · grading the checks

Any five specific claims: the birth year and place · the invention of the rotary seed drill · the 1923 patent · the figure of 4,000 farms · the 212 employees · the Dearborn Medal in 1948 · the street name.

Good checking methods: search a patent register · check the official list of medal recipients · look at a map or street directory · search a library or newspaper archive · look for any **independent** source that is not simply repeating the same paragraph.

Fastest to check: the street name — an online map answers it in seconds. Give full credit for any answer that picks the claim with a single, public, searchable record behind it.

Task 9.3 · where classes usually land

a. OK — the thinking and the writing are the student's. This is using AI as a tutor.
b. NOT OK — near-universal agreement. The work is not theirs.
c. OK — no different from a spellchecker, which nobody objects to.
d. OK — ideas are not the assessed part; the research is.
e. DEPENDS, and this is the best argument on the page. If the task was to *read* the book, a summary defeats it entirely. If the book was background for something else, it may be perfectly reasonable. Push the class to name **what was actually being assessed** — that is the principle underneath every one of these.

Task 9.4 · the enforcement question

Most student rules are unenforceable, and that is fine. An unenforceable rule can still be a good one if people agree with it — most school rules work that way. Accept any answer that separates **whether a rule is right** from **whether it can be policed**.

LESSON 10 · TEACHER BRIEFING

Your Digital Footprint

READ BEFORE THE BELL

2 MINUTES

Students examine a fictional student's public posts and work out exactly how much a stranger could learn from them. It is genuinely unsettling in a productive way, and it needs no technology at all.

You do not need to know this subject

Everything here is reading and judgment. The fictional profile is on the student page and the answer key lists every piece of information that can be pieced together from it.

Photocopy these pages only

Pages 52–54

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Do not ask students to bring up their own real accounts in class.

WHAT TO SAY TO START

> *"Nobody posts their address online. But six ordinary posts, none of them careless on its own, can add up to your address, your school, and the exact times your house is empty. Today you are going to prove that."*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the three-part panel together	The third part surprises them most
12–28	Task 10.1 — the profile audit	The key lists everything findable
28–40	Task 10.2 — rating information by risk	Expect argument about phone numbers
40–50	Task 10.3 — the future test	Answer key has guidance
50–55	Task 10.4, collect sheets	Discussion prompt below

Questions students will ask

"None of that is private though."
Correct — that is precisely the point. No single post is careless. The risk comes from combining them, which is what Task 10.1 makes them do.

"Should I delete my accounts?"
No, and avoid saying so. The lesson is about what you post and who can see it, not about withdrawing from the internet.

"Can you really find someone from a photo?"
Often, yes — uniforms, street signs, bus numbers and storefronts are all in the background of ordinary photos. Keep this general rather than demonstrating it.

If they finish early

Task 10.4 asks students to write advice for a 7th grader. Strong students can then argue the opposite case: what are the genuine costs of having no online presence at all when you apply for a job or a college place?

If the computers are working

Students can search their own name in a browser they are not signed in to, and see what comes back. Set clear boundaries first: their own name only, and no searching for classmates.

Curriculum focus · for the regular teacher's records

Digital footprints and personal data; privacy and security of personal information; responsible use of digital systems. Discussion prompt for the last five minutes: *"You can delete a post. Can you delete a screenshot somebody else took of it?"*

LESSON 10 · YOUR DIGITAL FOOTPRINT

Six harmless posts

UNDERSTAND

12 MIN

NAME

CLASS

DATE

What you post

Photos, comments, videos. The part you control — and the part people think is the whole picture.

What others post

Tagged photos, group chats, a friend naming your school. You did not choose any of it.

What is simply recorded

Locations, times, which device, what you searched. Nobody posted it. It exists anyway.

Public posts from one account · all visible to anyone

WHEN	THE POST	THE PHOTO
MON 7:52am	Bus is late AGAIN. Third time this week	a bus stop and a route 512 sign
MON 4:10pm	Practice til 6 every Monday and Wednesday. Dead.	a sports field and a school hoodie with a visible logo
WED 8:30pm	Happy birthday to my absolute legend of a mom!! 47 today	taken inside a kitchen
FRI 3:40pm	Anyone else at Northgate Plaza after school? Im at the food court	none
SAT 11:15am	Whole family away next week, cant wait	four packed suitcases in a hallway
SUN 9:00pm	New phone!! Finally	the box, and a school planner on the desk behind it with a name label

LESSON 10 · YOUR DIGITAL FOOTPRINT

What can a stranger work out?

KEY SKILL

18 MIN

Task 10.1 · The audit

INVESTIGATE

Using only the six posts on the previous page, work out what a stranger could learn. Write the post that gives it away.

WHAT A STRANGER COULD LEARN	WHICH POST GIVES IT AWAY?
Which school they attend	
Roughly which suburb they live in	
Their full name	
Two afternoons they are definitely not home	
A whole week the house will be empty	
Their mother's year of birth	
Something valuable that is in the house	
Where to find them on a Friday afternoon	

Which single post is the most dangerous, and why? _____

Task 10.2 · Rate the risk

JUDGE

Rate each one from 1 (harmless) to 5 (never post this).

INFORMATION	1–5	INFORMATION	1–5
Your first name		Your weekly routine	
Your school		When your house is empty	
Your suburb		A photo in school uniform	
Your date of birth		Your phone number	

Pick any two you rated low on their own. Explain how they become dangerous **together**.

The long memory

STRETCH · 16 MIN

Task 10.3 · Would this matter in eight years?

DECIDE

Imagine each of these is found by an employer, a college, or a team selector when you are 23. Mark it **fine**, **awkward** or **damaging**, and say why.

#	THE POST	VERDICT	WHY
a	A photo of you winning a school competition		
b	An argument in the comments where you insulted someone		
c	A joke about a teacher, using their real name		
d	A strong political opinion you no longer hold		
e	A photo of you at a party, tagged by someone else		

Task 10.4 · Advice for a 7th grader

EXTENSION

Write three pieces of advice for a 7th grader getting their first phone. Make them realistic — “never post anything” is advice nobody follows.

1. _____
2. _____
3. _____

Task 10.5 · The last word

THINK

You can delete a post. Explain in one sentence why that does not necessarily mean it is gone.

_____

_____

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

_____

_____

Teacher initials _____

Answer key

Your Digital Footprint

NOT FOR STUDENTS

ANSWERS

Task 10.1 · everything findable in six posts

WHAT A STRANGER LEARNS	FROM WHICH POST	HOW
The school	Monday 4:10pm	School logo visible on the hoodie
The suburb	Monday 7:52am + Friday	Bus route 512 and Northgate Plaza together narrow it right down
Their full name	Sunday 9:00pm	Name label on the school planner in the background
Two afternoons away	Monday 4:10pm	Practice every Monday and Wednesday until 6
A week the house is empty	Saturday 11:15am	“Whole family away next week” plus packed suitcases
Mother’s birth year	Wednesday 8:30pm	Age 47 and the date of the post
Something valuable inside	Sunday 9:00pm	A brand new phone, shown in its box
Friday whereabouts	Friday 3:40pm	Named the mall and the food court

The most dangerous post is Saturday. It publicly announces an empty house for a known week, to an audience that already knows the suburb from earlier posts. On its own it is harmless. Combined with the other five it is an invitation. That combination is the entire lesson.

Task 10.2 · guidance

Ratings are a judgment, not a key. Look for reasoning. The usual pattern:

Low on their own: first name, suburb, a school photo.

High on their own: phone number, when the house is empty, full date of birth.

The combination question is the one that matters. School + routine gives someone a time and a place to find a specific child. Neither piece is alarming alone.

Task 10.3 · what to draw out

a is fine and worth pointing out — a footprint can help you. **b** and **c** are damaging because they show how someone treats other people. **d** is the interesting one: people are allowed to change their minds, but the internet does not record that they did. **e** raises the real problem — **you did not post it and you cannot delete it**.

Task 10.4 and the closing question

Good advice looks like: check what a post shows in the background, not just the subject · wait ten minutes before posting when angry · never announce that you are away until you are home · assume a private post can be screenshotted · review who can see your account once a semester.

The closing question: deleting removes your copy, not everyone else’s. Screenshots, archived pages, and other people’s reposts all survive. Accept any answer reaching “someone else may already have a copy”.

AFTER THE LESSON

Handover note

LEAVE THIS ON THE DESK

90 SECONDS

Photocopy this page or fill it in directly. Ninety seconds of your time saves the regular teacher a great deal of guesswork.

SUBSTITUTE TEACHER	CLASS	DATE	PERIOD

Lesson used

CHECK ONE

- ☐ 01 · Follow My Instructions Exactly
- ☐ 02 · Counting Like a Computer
- ☐ 03 · Trace the Code
- ☐ 04 · Flowchart Detective
- ☐ 05 · How Attacks Actually Work
- ☐ 06 · Sending a Message Across the World
- ☐ 07 · Sorting and Searching by Hand
- ☐ 08 · Bug Hunt
- ☐ 09 · What AI Can and Cannot Do
- ☐ 10 · Your Digital Footprint

How far did the class get?

- ☐ Everyone finished the main tasks
- ☐ Most finished; some are partway
- ☐ We got through about half
- ☐ We did not get far — see notes

Sheets

- ☐ Collected and left on the desk
- ☐ Students kept them
- ☐ Graded against the answer key

Anything the regular teacher should know

NOTES

Students to mention · for any reason, good or otherwise

NAMES

Thank you

Walking into an unfamiliar subject is not easy. Leaving a clear note behind is the part that makes the next teacher’s day work, and it is genuinely appreciated.

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Curriculum alignment

CSTA K–12 Computer Science Standards, levels 2 and 3A. Across the ten lessons this volume touches:

- Designing and representing algorithms
- Representing data in binary and other forms
- Tracing and desk-checking programs
- Branching, iteration and logic errors
- Cyber security, passwords and social engineering
- How networks transmit and route data
- Algorithm efficiency: sorting and searching
- Testing, debugging and error types
- Artificial intelligence, evidence and ethics
- Digital footprints, privacy and personal data

Individual lesson teacher pages name the specific focus for that lesson.

The rest of the series

Volume 2 · Data, Systems and Design
Data cleaning, misleading charts, databases, input-process-output, compression, encryption, interface design, functions, project planning and the cost of free apps.

Volume 3 · Machines and Meaning
Lists, Boolean logic, hardware, files, requirements, machine learning, DNS, search ranking, control systems and copyright.

Volume 4 · Media, Evidence and Impact
Images and sound as data, stacks and queues, digital forensics, version control, network design, accessibility, algorithmic decisions, IoT security and automation.

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Physical Computing with the Raspberry Pi 5 · AI Literacy for Secondary Students · and more.

Found an error, or want a lesson on something else?

Feedback is genuinely read. If a lesson did not land with your class, or you need a sub plan on a topic that is not here, get in touch through digitalvector.com.au and it will be considered for the next update.

EMERGENCY SUB PLANS PACK

Digital Technologies · Grades 7–10

Volume One · ten lessons · no preparation · no computers required